BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

Cluster Programme - M. Tech in AI & ML

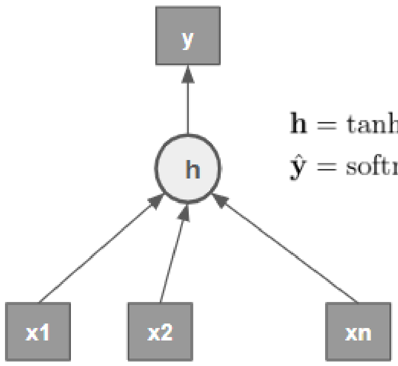
II Semester , 2023 – 24(July,2024)

COMPREHENSIVE Examination (**Regular**)_ANSWER KEY

Course No	:	AIMLC ZG511
Course Title	:	DEEP NEURAL NETWORK
Nature of Exam.	:	Open Book
Weightage	:	35 Marks
Duration	:	120 minutes
Date	:	29th September,2024_2 PM

Number of questions:4
Number of Pages: 2

Q.1.a). Consider the following neural network architecture with one hidden unit and one output unit as follows: **[4 M]**



$$\mathbf{h} = \tanh(W\mathbf{x} + \mathbf{b})$$
$$\hat{\mathbf{y}} = \mathrm{softmax}(U\mathbf{h} + \mathbf{d})$$

Assume a sequence length of size ***n*** with each xi represented by a ***m***-dimensional vector. Let ***h*** represent a single hidden layer with ***H*** hidden units and ***tanh*** activation. Model predicts y using ***softmax*** function with U representing the weight matrix and d representing the bias for node **y**.

i) What are the dimensions of W, U, b, h, d and the input matrix X.

Solution 1a) Ans i): **The dimensions of W, U, b, h, d and the input matrix X are: [2 M]**

X : n × m (sequence of length , each vector of dimension).

W: m × H (weight matrix from input to hidden layer).

b: H × 1 (bias for hidden layer).

h: n × H (hidden layer output for the sequence).

U: H × c (weight matrix from hidden layer to output layer, where is the number of output classes).

d: c × 1(bias for output layer).

ii) What are the total number of parameters in the above architecture?

W has $m \times H$ parameters.

b has H parameters.

U has $H \times c$ parameters.

d has c parameters.

Thus, the total number of parameters is $(m \times H) + H + (H \times c) + c$.

b).Two friends Mr Raju and Mr Robert, while experimenting with a deep neural network, found that their model is over-fitting the given training data. The friends referred to the available literature and materials and came up with a solution to reduce the over-fitting. Mr Raju decided to add 1 drop out along with batch normalization to his model. Whereas Mr Robert decided to go with a deeper model, hence he added more layers. But he found that his model is struggling to train because of his hardware limitations. So he decided to employ early stopping. Assess whether the decisions Mr Raju and Mr Ravi are taking helps them to proceed in the right direction. **[4 M]**

Solution :In assessing the decisions made by Mr. Raju and Mr. Robert it appears that Mr. Raju has adopted a sound strategy by implementing dropout and batch normalization to address overfitting, while Mr. Robert's approach may lead to complications. Additionally, he has correctly identified early stopping as a useful technique to manage training without exhausting resources, which is a positive adjustment.

Mr. Raju's Approach: Dropout and Batch Normalization [1.5M]
Mr. Raju's implementation of dropout is a recognized method for reducing overfitting in neural networks. Dropout works by randomly disabling a fraction of neurons during training, which helps to prevent co-adaptation among neurons, thus promoting better generalization. The addition of batch normalization complements this by normalizing the inputs to each layer, which stabilizes the learning process and speeds up convergence, further aiding the model's performance. This dual approach is indeed effective for enhancing the robustness of the neural network against overfitting.

Mr. Robert's Approach: Deeper Model and Early Stopping [1.5M]
In contrast, Mr. Robert's decision to deepen his model by adding more layers can exacerbate the issue of overfitting. A deeper model has the capacity to memorize the training data rather than generalizing from it, particularly if the amount of training data is insufficient. Hence, increasing model complexity without adequate regularization methods can lead to poor generalization on unseen data.

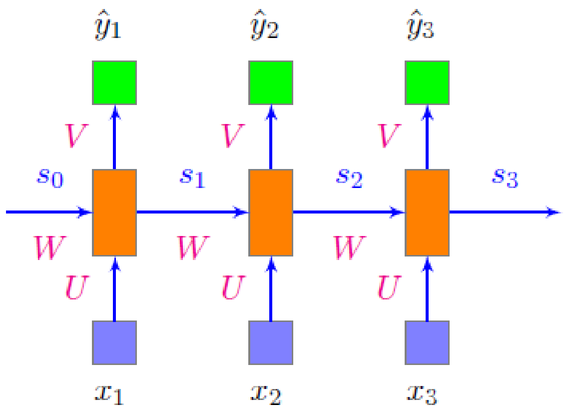
Recognizing hardware limitations, Mr. Robert's decision to incorporate early stopping is prudent. Early stopping serves as a regularization technique by halting training when the model's performance on the validation dataset begins to decline, thus preventing unnecessary overfitting. This technique helps to conserve computational resources while ensuring that the model does not overfit the training data. Nevertheless, the root issue of increasing model complexity remains a concern, and early stopping alone may not suffice to achieve the best generalization.

Final Assessment[1M]

Ultimately, Mr. Raju is on the right path with his combination of dropout and batch normalization, both effective methods for combating overfitting. Mr. Robert's strategy has merits, particularly with the inclusion of early stopping, but he may still face challenges due to the detrimental nature of a more complex model. A better approach for Mr. Robert might involve reassessing the architecture of his model to avoid unnecessary complexity while still implementing regularization techniques like dropout and leveraging early stopping.

Q.2.a) Compute the outputs in each timestep and the state after timestep = 3 for the Vanilla RNN given below. Assume the biases as zeros. **[8 M]**

$X = [\,1\ 1\ 0]^T$ $W = [\,0.2\ \ 0.3\ \ 0.8]^T$ $U = [\,0.5\ \ 0.6\ \ 0.2]^T$ $V = [0.4\ \ 0.2\ \ 0.1]^T$



(a) Compute the outputs in each timestep and the state after timestep=3 for the Vanilla RNN given below. Assume the biases as zeros. [5]

$$X = \begin{bmatrix}1 & 1 & 0\end{bmatrix}^\top$$
$$W = \begin{bmatrix}0.2 & 0.3 & 0.8\end{bmatrix}^\top$$
$$U = \begin{bmatrix}0.5 & 0.6 & 0.2\end{bmatrix}^\top$$
$$V = \begin{bmatrix}0.4 & 0.2 & 0.1\end{bmatrix}^\top$$
$$s_t = \sigma(Ux_t + Ws_{t-1} + b)$$
$$\hat{y}_t = Relu(Vs_t + c) \qquad \text{Relu is assumed}$$
$$s_0 = 0 \qquad b = 0 \qquad c = 0$$
$$s_1 = \sigma(0.5 * 1 + 0.2 * 0 + 0) = 0.6$$
$$\hat{y}_1 = \max(0, 0.4 * 0.6 + 0) = 0.24$$
$$s_2 = \sigma(0.6 * 1 + 0.3 * 0.6 + 0) = 0.68$$
$$\hat{y}_2 = \max(0, 0.2 * 0.68 + 0) = 1.36$$
$$s_3 = \sigma(0.2 * 0 + 0.8 * 0.68 + 0) = 0.63$$
$$\hat{y}_3 = \max(0, 0.1 * 0.63 + 0) = 1.26$$

b) A company is trying to automate generating reports from videos of ultrasound scans. The videos of a beating heart of fetus at a particular section or slice. The report to be generated describes the heart condition and its location and any abnormalities detected. The videos are of 36 frames each. Each frame is 360 x 240 grayscale image, **[4 M]**

a) Suggest suitable type of RNN to be used for this application and justify your selection.

b) Compute the depth of the RNN suggested

c) Should an LSTM or GRU or Bidirectional RNN should be used for this application. Justify your choice.

(b) (i) Many to many RNN has to be used. For each frame, the position of the fetus and the condition has to output. For each frame / input, there is an output.
(ii) Depth = 36, the number of frames per video.
(iii) LSTM is better.

Q.3. Consider a text sequence with 3 words : <w1, w2, w3> such that each word is represented by a 2-d vector as given in matrix X. Show step by step calculations for computing the scaled dot-product based self attention scores using the Wq, Wk and Wv matrices given below**[6 M]**

W_q		
2	0	2
2	0	0
2	1	2

W_k		
2	2	2
0	2	1
0	1	1

W_v		
1	1	0
0	1	1
0	0	0

ANSWER

Assuming matrix X is given as follows (Student need to assume X and complete it accordingly, check carefully)

X = 1 0 0

0 1 0

0 1 1

Q = X * Wq

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 0 & 2 \\ 2 & 0 & 0 \\ 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 \\ 2 & 0 & 0 \\ 4 & 0 & 2 \end{bmatrix}$$

K = X * Wk

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 2 \\ 0 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 0 & 2 & 1 \\ 0 & 3 & 2 \end{bmatrix}$$

V = X * Wv

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Award 2 marks for each correct Matrix

b).i) Mention and justify the different types of attention mechanisms used in Transformer architecture. **[2 M]**

ANSWER:

1. Self-Attention

used by both encoder and decoder stacks to compute the attention scores for better contextual representation of te input sentence (for encoder) adn sequence generated thus far (decoder)

2. Masked attention or Cross Attention

This is used only in the decoder stack so that we can compute attention scores between the encoder and decoder to generate the next token

Award 1 mark to write both the names correctly and 1 mark for the justification

ii) Justify the selection of Bahdanau attention and disadvantages due to this selection. Is there any alternative to this attention?

[2 M]

ANSWER

Bahdanau Attention provides the mechanism to focus on specific parts of the input segment while generating parts of the output sequence. At each time step of the decoder, we compute a weighted sum of attention scores from the hidden states of the encoder.

Disadvantgae :

1) Due to computation of attention weights at each stage in a sequential manner, process is slow and cannot be parallelized.

2) Computing attention scores is quadratic time complexity with respect to input sequence length.

Award 1/2 mark for any of the above mentioned disadvantages

Self-attention as used in Transformers is an alternative way to compute attention scores.

Or Luong attention can is an alternative where the computation is less

[1 M]

Q.4. a) A medical doctor has approached you to help her identify a rare disease from MRI images.

The doctor provides you with a lot of MRI images and 75% of them have the region of interest marked or annotated. You decided to regress the region of interest from the images using a suitable network. Identify and justify. **[2.5 M]**

i) Identify the neural network that you will use to determine suspicious / abnormal behavioral patterns in the computer networking of an organization

ii) A combination of CNNs and RNNs are used to provide a description of an image. Justify how CNN and RNN are used.

Answer:

Convolutional Neural Networks (CNNs) are best suited for image segmentation i.e. to train a model that can accurately locate and outline ("segment") the region of interest (ROI) in new MRI images. Transfer Learning – usage of pre-trained CNNs (like ResNet, Inception, or VGG) that have already learned valuable features from massive image datasets.

Justifications:

- CNNs are good at Feature Extraction - automatically learn and extract features from images, which is crucial for identifying the ROI in MRI scans.
- Spatial Invariance and Efficiency

i. Identify the neural network that you will use to determine suspicious / abnormal behavioral patterns in the computer networking of an organization

Answer:

For detecting suspicious or abnormal behavioral patterns in computer networking, Recurrent Neural Network (RNN), particularly a Long Short-Term Memory (LSTM) network, shall be used.

ii. A combination of CNNs and RNNs are used to provide a description of an image. Justify how CNN and RNN are used.

Answer:

CNNs for Feature Extraction – to extract features (edges, textures, shapes, and other relevant features) from the image.

RNNs for Sequence Generation - the extracted features are fed into an RNN (or LSTM) to generate a sequence of words that describe the image.

b) A research group is working on a project related to medical images classification / clustering. For this they require medical images from diversified super specialty hospitals so that the model can be trained on diversified images so that more generalized model can be obtained. But the research group is facing difficulties in obtaining images due to various regulatory guidelines and also they are not having sufficient computational infrastructure. Suggest a suitable and feasible solution to the group. **[2.5 M]**

Answer:

Federated learning enables model training on decentralized datasets. This helps in overcoming both the difficulties:

1. Difficulty in obtaining images: Instead of moving data to a central server, the model is sent to each hospital. Each hospital trains the model locally on its own data and only model updates (e.g., gradients) are sent back to the server / research group. The server aggregates updates and sends an improved model back to the hospitals.
2. Insufficient computational infrastructure: In Federated learning, the computational load is distributed across multiple hospitals, reducing the need for a single, powerful computational infrastructure. Hospitals can use their existing computational

resources (like edge devices) to participate in the training process, minimizing the load in the server.

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